

Metallographic Microanalysis of Ferrous and Copper Alloys

Identifying six unknown specimens from their microstructures using optical and digital microscopy

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Course	ME 215-102, NJIT	Instructor	Prof. Suwattananont
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PROJECT SUMMARY

This project used microanalysis to identify six coded metal specimens from their microstructures. Each specimen was examined and photographed with a conventional light microscope and then a digital microscope, which produced clearer images for identification. An image analyzer measured the size and amount of phases, and reference mechanical properties were compiled to connect each structure to its behavior. With confirmation from the instructor and TA, the specimens were identified as aluminum bronze, hypoeutectoid steel, hypereutectoid steel, nodular cast iron, gray cast iron, and white cast iron. Results were largely consistent with expected microstructures, with small deviations traced to an unrepresentative viewing area and insufficient magnification.

Keywords: microanalysis, hypoeutectoid steel, hypereutectoid steel, aluminum bronze, nodular cast iron, gray cast iron, white cast iron

1. Introduction & Objectives

A material’s performance in any application depends on its microstructure — the arrangement of grains and phases visible only under magnification. Microscopy has transformed materials engineering by making these details observable. The objectives of this project were to learn how specimens are prepared for microscopy, operate both conventional and digital light microscopes, analyze images to identify microstructures, and determine the size and amount of phases within an alloy in order to understand properties that are not visible to the eye.

2. Background

Because a material’s microstructure directly controls its properties and suitability for an application, microanalysis is central to any thorough material evaluation. Advances in microstructural characterization, especially digital image capture and processing, have expanded what engineers can observe and measure, and better material understanding leads directly to more efficient and durable products (Brandon & Kaplan, 2013).

3. Procedure

The session began with an instructional video on microanalysis. Six coded specimens were provided, representing hypoeutectoid steel, hypereutectoid steel, aluminum bronze, gray cast iron, nodular cast iron, and white cast iron. Each specimen was first examined under a standard light microscope, adjusting focus and magnification, and photographed. The specimens were then re-examined with a digital microscope to improve image quality and identification accuracy. All micrographs were compared against reference images and recorded with their magnification. Finally, an image analyzer was used to measure the size and amount of phases, and results were reviewed and signed off by the TA.

4. Results

Table 1. Specimen identification (confirmed by instructor and TA)

Sample ID	Objective	Identified material	Key features observed
E8	M5	Aluminum bronze	Pale α phase with dark β / intermetallic regions
N15	M5	Hypoeutectoid steel	Ferrite with pearlite colonies
N20	M5	Hypereutectoid steel	Pearlite with cementite network
N62	M5	Nodular cast iron	Dark spherical graphite nodules
M6	M20	Gray cast iron	Graphite flakes in pearlite/ferrite matrix
M12	M5	White cast iron	Cementite with pearlitic regions

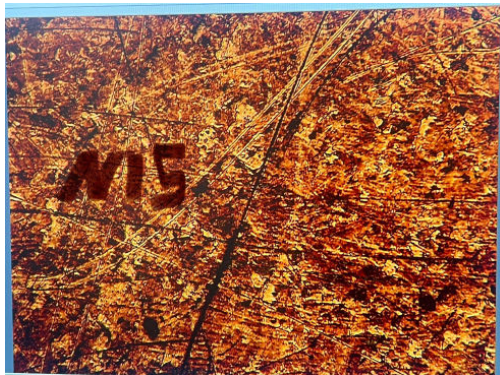


Figure 1. Specimen N15 — hypoeutectoid steel.

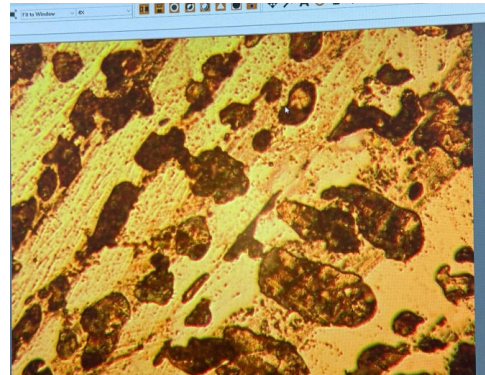


Figure 2. Specimen M12 — white cast iron.

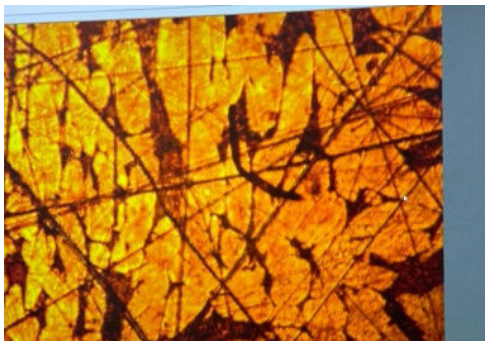


Figure 3. Specimen E8 — aluminum bronze.

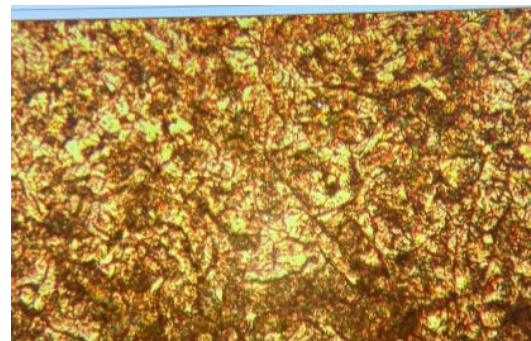


Figure 4. Specimen M6 — gray cast iron.

Finding the best magnification for a clean image took trial and error; some early images lacked sharpness, which made comparison harder. The digital microscope's color contrast made structures far easier to distinguish, and

every specimen could ultimately be identified. In the final step, the TA ran the aluminum bronze specimen through the image analyzer, which displayed the microstructure and measured feature lengths and areas.

Table 2. Reference mechanical property ranges compiled for each material

Material	Ultimate tensile (psi)	Yield (psi)	Elongation (%)	Hardness (HB)	Elastic modulus (GPa)	Machinability (%)
Hypereutectoid steel	2.33E4–4.64E5	3.49E4–4.84E5	0.5–30	165–300	15–235	30–40
Hypoeutectoid steel	6.38E4	5.37E4	15	85–550	183–213	65–75
Aluminum bronze	7.47E4	2.97E4	12	170–195	120–125	20
Nodular cast iron	6.5E4	4.5E4	12	120–600	65–170	60–70
White cast iron	5E4–9.99E4	7.01E4	1–10	350–810	165–250	10–125
Gray cast iron	1.67E4–1.02E5	9.5E3–2.5E4	1–15	120–550	60–160	0–125

5. Discussion

Hypereutectoid steel showed dark regions indicating cementite (Fe_3C) along with fine gray pearlite. At the magnification used, the excess cementite did not show the expected spheroidal shape; this morphology became visible at a different magnification and matched the reference images provided by the TA.

Hypoeutectoid steel consisted mainly of ferrite and pearlite, as expected. The sample showed more ferrite relative to pearlite than typical, likely because the microscope was focused on an unrepresentative area; several additional areas were examined to give a more accurate assessment.

Aluminum bronze showed a pale yellow α phase (aluminum dissolved in copper) with dark β -phase regions formed by intermetallic compounds. This matches the expected microstructure with no notable deviations.

Nodular cast iron showed dark spherical graphite nodules, formed when magnesium is added as a nodulizer. The structure was free of defects and characteristic of nodular iron.

White cast iron showed dark pearlitic regions with white cementite, including alternating zones of ferrite and cementite, consistent with its expected structure.

Gray cast iron showed dark graphite flakes, with darker pearlite and lighter ferrite regions in the matrix, consistent with known characteristics and no significant deviations.

Subtle variations in microstructure were sometimes difficult to detect, and scratches and other surface flaws obscured parts of several images. With guidance from the instructor and TA, all identifications were confirmed. Most data supported the objectives, while the small variances suggest that better sample preparation and more careful magnification selection would improve results.

6. Conclusion

This project successfully classified six materials by their microstructures and demonstrated the full microanalysis process. Aluminum bronze was identified by its distinct α and β phases, and nodular cast iron by its graphite nodules. Two deviations stood out: the unexpectedly high ferrite fraction in hypoeutectoid steel, likely from an unrepresentative viewing area, and the absence of spheroidal cementite in hypereutectoid steel, likely from insufficient magnification. Both highlight the importance of representative sampling and appropriate magnification. Future work could improve accuracy by refining magnification settings and examining more locations on each specimen. Overall, the project showed how microscopy connects microscopic structure to engineering properties.

7. Key Concepts

What are the main constituents of a medium-carbon alloy steel at room temperature?

Iron as the base metal, with carbon and manganese as the key alloying elements; microstructurally, ferrite and pearlite.

Does pearlite contain cementite?

Yes. Pearlite is a lamellar (layered) mixture of about 88% ferrite and 12% cementite.

How does grain size affect mechanical properties?

Larger grains generally give lower hardness and tensile strength but higher ductility. Smaller grains give higher hardness and strength, with lower ductility.

What is the composition of AISI 4340 steel, and what properties matter in an aircraft member?

Nominally 0.38–0.43% C, 0.60–0.80% Mn, 0.15–0.30% Si, 1.65–2.00% Ni, 0.70–0.90% Cr, 0.20–0.30% Mo, with P and S kept low ($\leq 0.035\%$ and $\leq 0.040\%$) and the balance iron. For aircraft components, high strength, toughness, and fatigue resistance are the key properties.

Can plain-carbon 1025 steel be welded?

Yes. With its low carbon content, 1025 steel has good weldability and can be joined by most common welding processes.

What is the difference between bronze and brass?

Both are copper alloys. Bronze is primarily copper and tin; it is strong, wear-resistant, and corrosion-resistant. Brass is primarily copper and zinc; it is easier to machine and form and generally has higher electrical conductivity than bronze.

What are the common types of cast iron?

White, gray, nodular (ductile), malleable, and compacted graphite iron.

Is aluminum known for excellent thermal conductivity?

Yes. It also offers an excellent strength-to-weight ratio and good formability.

How do gray and nodular cast iron differ in microstructure?

In gray iron, graphite forms as interconnected flakes, which act as stress concentrators. In nodular iron, magnesium makes the graphite form as spheres, which gives nodular iron much higher strength, ductility, and elongation.

References

Brandon, D., & Kaplan, W. D. (2013). *Microstructural characterization of materials* (2nd ed.). Wiley.